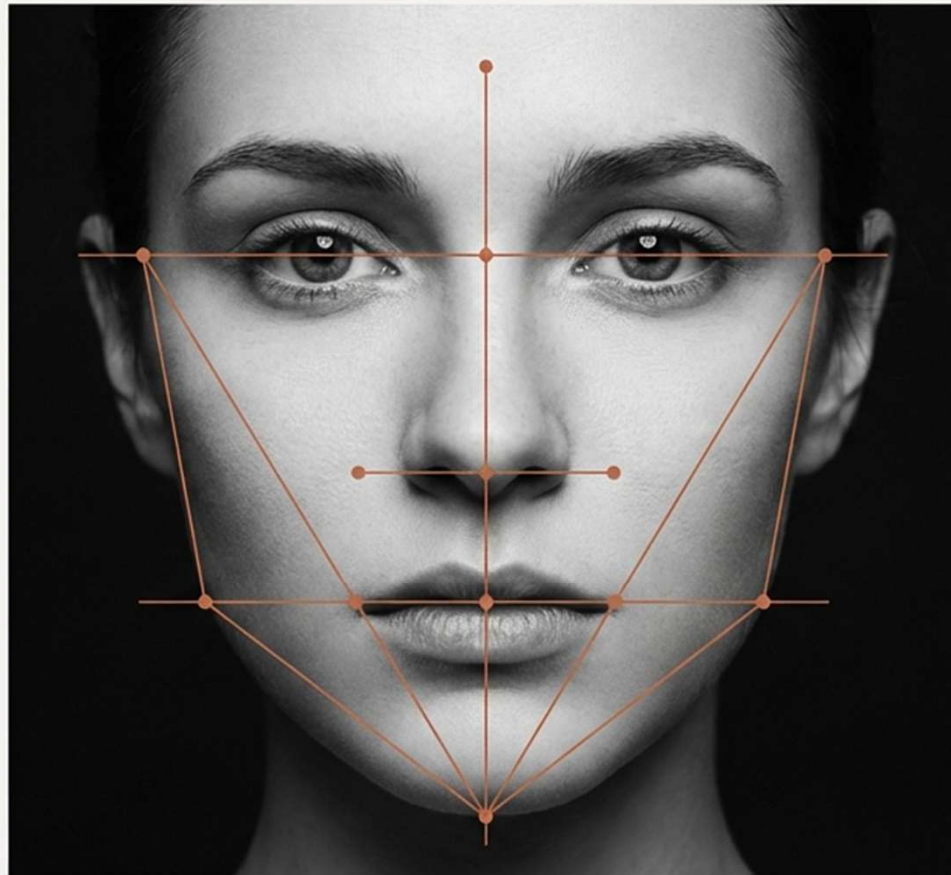


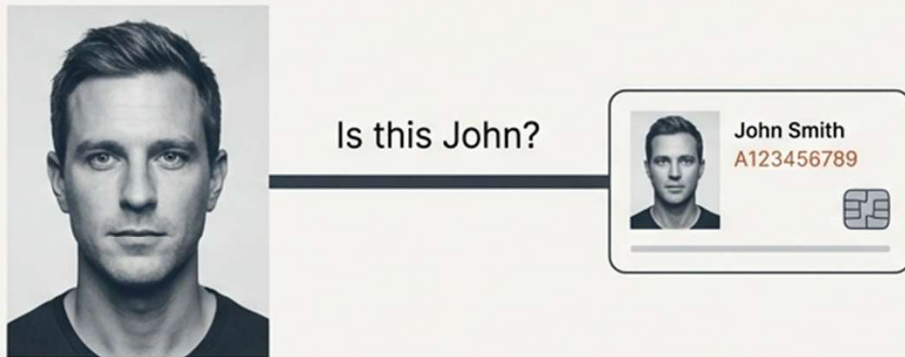
Deep Learning for Face Recognition



The distinction between verification and recognition

Face Verification (1:1 Matching)

Given an input <image, name/ID>, output whether the input image is of the claimed person.



Face Recognition (1:N Matching)

Given an input image and a set of 'n' stored images, determine whether the input image matches with any of the stored images.

Who is this among N people?



The constraint of One-Shot Learning



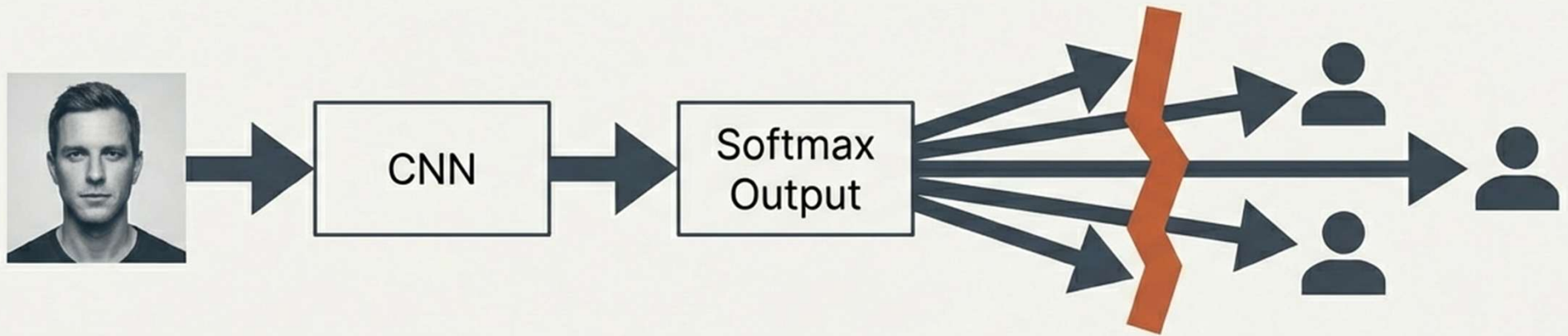
In most real-world cases, we only have a single image to learn and recognize a person's face.

Database



Traditional Deep Learning requires thousands of examples. With only one existing image per person, standard algorithms collapse when a new person is introduced.

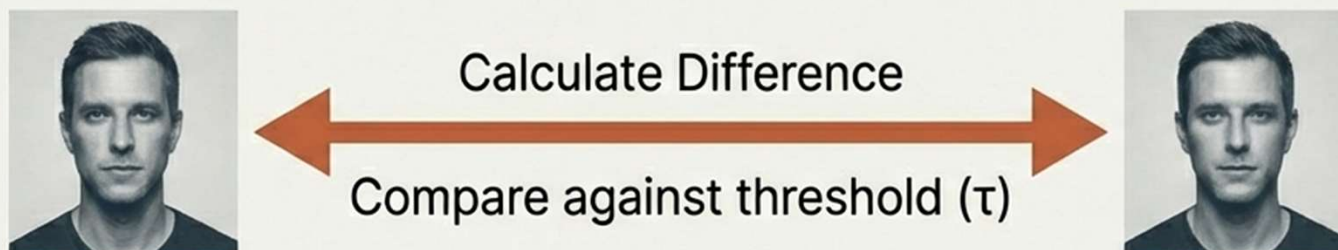
Why standard classification models fail



- 1. Robustness decreases significantly as more persons are added to the database.**
- 2. We must completely retrain the entire neural network every single time a new person is added.**

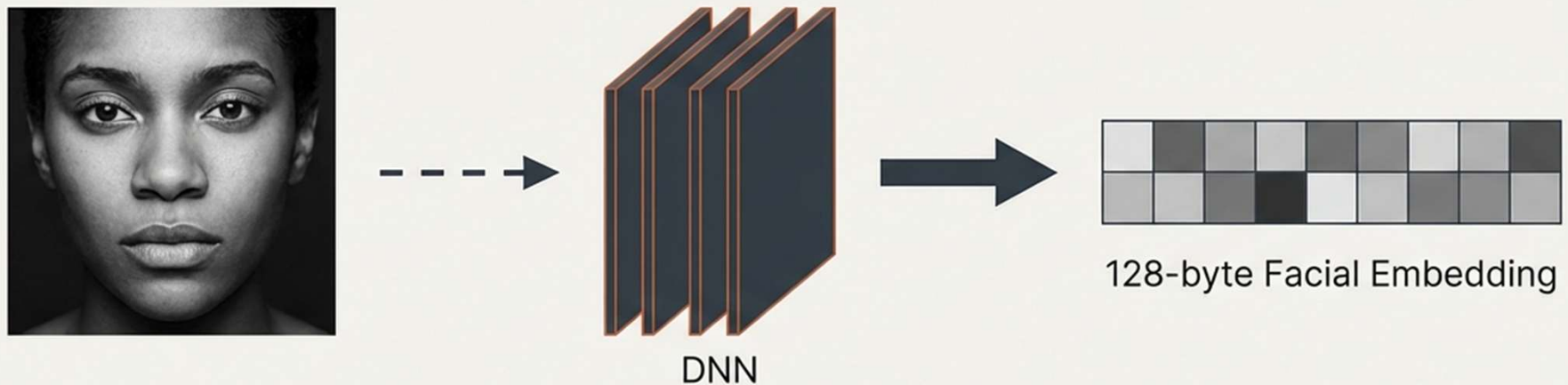
Shifting from classification to comparison

Don't Output a Label.
Output the Degree of Difference.



High output value = Different persons.
Output less than threshold (τ) = Verified match.

Distilling faces into mathematical encodings



The true job of the network is **compression**. Similar faces will result in similar **128-byte** encodings, while completely **different** faces will yield **highly distinct** mathematical representations.

Measuring the distance between encodings

Translating facial similarity into spatial distance.

Assume $x(i)$ and $x(j)$ represent the 128-byte encodings of two faces.

- If $x(i)$ and $x(j)$ are the same person, distance must be small.
- If $x(i)$ and $x(j)$ are different persons, distance must be large.

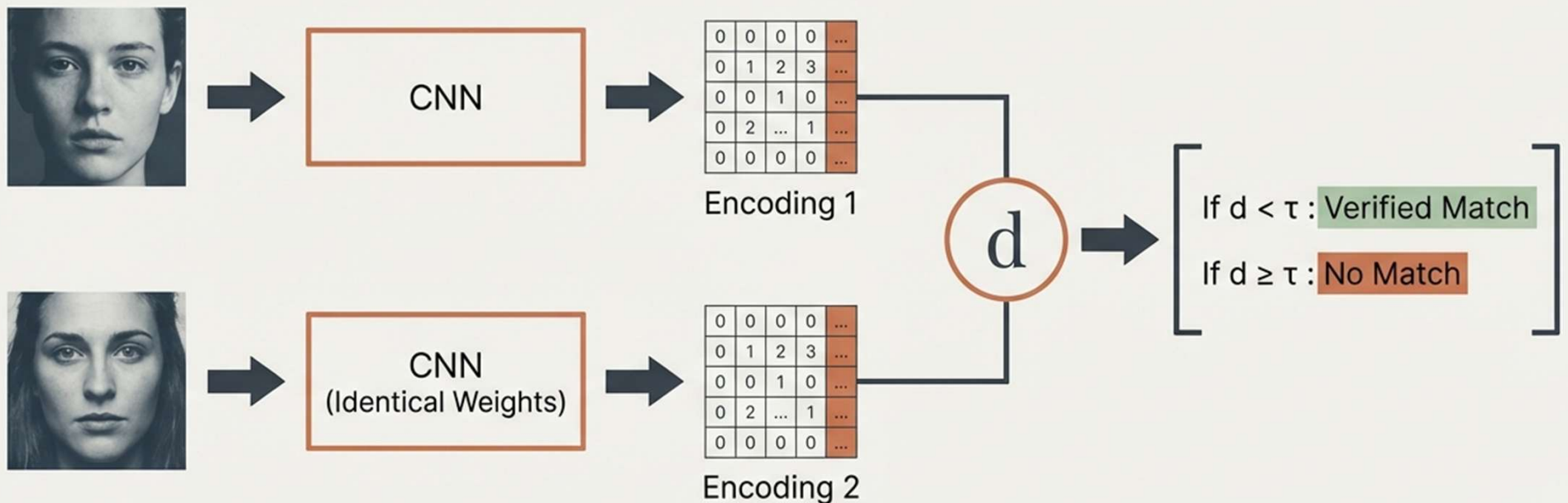
Parameters of NN define an encoding $f(x(i))$

Learn parameters so that:

If $x(i), x(j)$ are the same person,
 $||f(x(i)) - f(x(j))||^2$ is small

If $x(i), x(j)$ are different persons,
 $||f(x(i)) - f(x(j))||^2$ is large

Parallel processing with Siamese Networks



Solves the One-Shot inference problem without retraining.

The training dilemma

How do we teach the network to create these perfectly spaced encodings?

Inference:
One-Shot



Training:
Multiple Instances



To properly learn the mathematical mappings, training inherently requires multiple instances of a face (e.g., $m=10$).

Establishing the Triplet

Training the Siamese Network requires three distinct image types.



Anchor (A)

The baseline reference image.



Positive (P)

A different image, but of the same person.



Negative (N)

An image of a completely different person.

The spatial tug-of-war

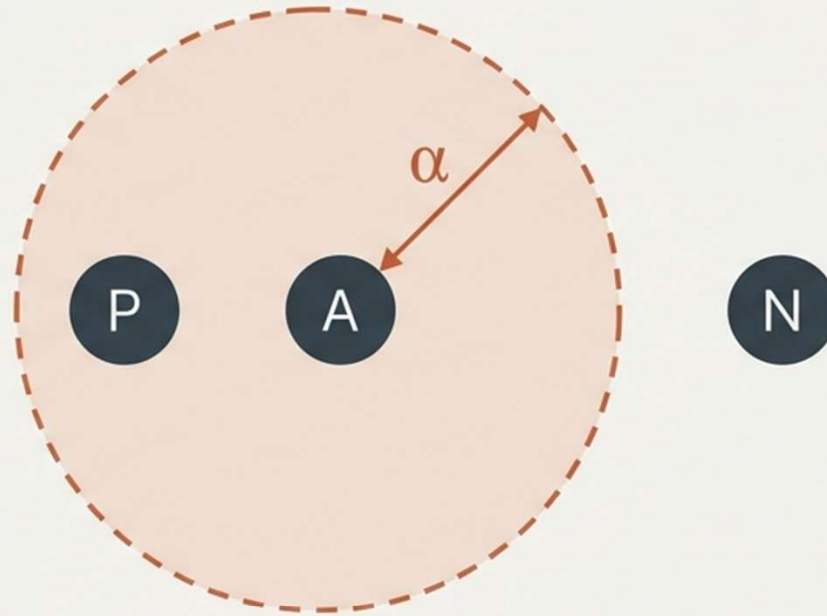


$$d(A,P) \leq d(A,N)$$

$$\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 \leq 0$$

Enforcing a definitive margin

It is not enough for the Positive to be simply “slightly closer” than the Negative. We need a definitive, significant gap to prevent lazy optimization.



$$\|f(A) - f(P)\|^2 + \alpha \leq \|f(A) - f(N)\|^2$$

The Triplet Loss Equation

$$L(A,P,N) = \text{Max}(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0)$$

If the gap term is greater than 0, it is selected and the network is penalized.

If the distance separation is already satisfactory, 0 is selected and no penalty is applied.

$$J = \sum L(A_i, P_i, N_i) \text{ for } i=1 \text{ to } m$$

Intelligent triplet selection

We cannot pick Anchor, Positive, and Negative images randomly. Random negatives are too easy for the network to distinguish.

We must intelligently select Hard Negatives where:

$$d(A,P) \approx d(A,N)$$




$$d(A,P) \approx d(A,N)$$

Keeping this initial difference low forces the network to learn and adapt based on very minor, nuanced facial differences.

Train on many, deploy for one

The Engine

Triplet Loss & Hard Negative Mining

- Requires training on multiple images (m)
- Learns the 128-byte mathematical encodings
- Enforces the margin parameter (α)

The Architecture

Siamese Networks

- Deployed in the real world
- Performs highly robust, One-Shot comparisons
- Calculates distance against threshold (τ) without retraining